Java Programming: Unit 4 - Control Flow & Iteration

Page 1: Foundations of Repetition
Loops in Java allow execution of a block of code multiple times based on a boolean condition.
The `while` loop checks its condition prior to executing the loop body:

```
while (condition) {
    // statement body
}
```

If the condition is initially false, the loop body will not execute even once.
Pre-test loops require careful variable initialization before the loop header.
Common pitfalls include omitting variable updates inside the loop, resulting in infinite execution.

Page 2: Do-While Loops & Comparison with While Loops

Unlike standard while loops, the `do-while` loop is a post-test loop.

Syntax:

```
do {
    // body executes at least once
} while (condition);
```

Key rule: The loop body is guaranteed to execute at least once, even if the condition is false on the first check.

Use do-while loops primarily for input validation where user interaction must occur prior to validation.

Page 3: For Loops, Invariants, and Off-by-One Boundaries
The `for` statement standardizes loop headers:
    for (initialization; condition; update) { ... }

Boundary Conditions and the Off-by-One Error:
A loop running `for (int i = 0; i <= n; i++)` iterates `n + 1` times when starting from 0.
In contrast, `for (int i = 0; i < n; i++)` iterates exactly `n` times.
When traversing zero-indexed structures, using `<=` rather than `<` causes an out-of-bounds error.
Carefully trace boundary conditions: verify whether the loop should terminate strictly before or at the bound.